PMM2
RTS Commands Interface

Revision 0.1.1

11/13/2020

Contents

RTS Commands 3
Range Tables 7
Voltage Ranges 7
Current Ranges (CH1-3) 7
Current Ranges (CH4) 7

RTS Commands

Command	**“DEVID<Family>:<Serial Number>:<Model>:<Hardware Rev>:<DSP Revision>;”**
Name	Device ID
Parameters	Family<PMM>:Serial Number<yyyymmddxxxx>:Model<2>:Hardware Rev<Int>:DSP revision<Int>
Behavior	Set PMM2 Device ID and save in Flash
Return	“DEVID;”
Comment	

Command	**“QC;”**
Name	Query Configuration
Parameters	-
Behavior	Returns the PMM2 configuration
Return	“Serial Number, Loader Rev, App Rev, CPLD Rev, Hardware Rev, Line Freq, Boot Count, DSP Rev, Input Voltage;”
Comment	

Command	**“QIP;”**
Name	Query IP
Parameters	-
Behavior	Returns the PMM2 IP address
Return	“IP,IP Address String;”
Comment	

Command	**“SU;”**
Name	Software Reset
Parameters	-
Behavior	Executes a Software reset
Return	“SU;”
Comment	

Command	**“SWI;”**
Name	Switch to Boot Loader
Parameters	-
Behavior	Reboots the DSP in Boot Loader mode
Return	“SWI;”

Comment	

Command	**“U;”**
Name	Unit Reset
Parameters	-
Behavior	Reboots the DSP
Return	
Comment	

Command	**“KILL;”**
Name	Kill
Parameters	-
Behavior	Reboot the DSP to Boot Loader Mode permanently
Return	“KILL;”
Comment	

Command	**“AUTORNG:<Flag>;”**
Name	Auto Range
Parameters	Flag<1-Enable | 0 Disable>
Behavior	Enable | Disable AUTO Range
Return	“AUTORNG;”
Comment	

Command	**“RANGE<Channel>:<Range>:<Flag>;”**
Name	Set Range
Parameters	Channel<1-4>:Range<See Table>:Flag<1-VPMM | 0 IPMM>
Behavior	Set the range for the specified channel
Return	“RANGE;”
Comment	

Command	**“RANGES<Range1>:<Range2>:<Range3>:<Range4>:<Flag>;”**
Name	Set Ranges
Parameters	Range1<Int>:Range2<Int>:Range3<Int>:Range4<Int>:Flag<1-VPMM | 0 IPMM>
Behavior	Sets the ranges for all the channels of the VPMM or IPMM
Return	“RANGES;”
Comment	

Command	**“RNGALL<Range>:<Flag>;”**
Name	Set Ranges
Parameters	Range<Int>:Flag<1-VPMM | 0 IPMM>
Behavior	Set all VPMM Channels to the same range
Return	“RNGALL;”
Comment	IPMM is not supported

Command	**“QVR;”**
Name	Query Voltage Ranges
Parameters	
Behavior	Returns the Active Voltage Ranges
Return	“VCH1-RA<Int>,PG<Float>,AG<Float>,NG<Float>…. VCH4-RA<Int>,PG<Float>,AG<Float>,NG<Float> ;”
Comment	RA=Range, PG=Prior gain, AG=Active Gain, NG=Next Gain

Command	**“QIR;”**
Name	Query Current Ranges
Parameters	
Behavior	Returns the Active Current Ranges
Return	“ICH1-RA<Int>,PG<Float>,AG<Float>,NG<Float>…. ICH4-RA<Int>,PG<Float>,AG<Float>,NG<Float> ;”
Comment	RA=Range, PG=Prior gain, AG=Active Gain, NG=Next Gain

Command	**“QMAC;”**
Name	Query MAC
Parameters	-
Behavior	Returns the MAC address of the DSP
Return	“M,<MAC Address Hex>;”
Comment	

Command	**“QT;”**
Name	Query Time
Parameters	-
Behavior	Returns the PMM2 Timer Time
Return	“TIME<Float>;”
Comment	

Command	**“QRYALL;”**
Name	Query All
Parameters	-
Behavior	Returns information on all measurements
Return	"V1ACRms,V1DCRms,V1Freq,V1Ph,V1Gain,V2ACRms,V2DCRms,V2Freq,V2Ph,V2Gain, V3ACRms,V3DCRms,V3Freq,V3Ph,V3Gain,V4ACRms,V4DCRms,V4Freq,V4Ph,V4Gain, VSampleId,V_SPI_CRC,Vnotused,Vnotused1, I1ACRms,I1DCRms,I1Freq,I1Ph,I1Gain,I2ACRms,I2DCRms,I2Freq,I2Ph,I2Gain, I3ACRms,I3DCRms,I3Freq,I3Ph,I3Gain,I4ACRms,I4DCRms,I4Freq,I4Ph,I4Gain, ISampleId,I_SPI_CRC,Inotused,Inotused1;"
Comment	

Command	**“GETHARM<Range>:<Flag>;”**
Name	Get Harmonics
Parameters	Channel<Int>:Flag<1-VPMM | 0 IPMM>
Behavior	Get Harmonics’ Amplitudes and listed as multiples of base frequency up to 50;
Return	“Amplitude at base freq,Amplitude at base freq*2,Amplitude at base freq*3…;”
Comment	

Range Tables

Voltage Ranges

Range	Range ID	Command	Gain
1	2V	0x1F	2.05
2	10V	0x17	10.3
3	30V	0x15	36.5
4	150V	0x0B	155.34
5	300V	0x0D	302.4
6	1000V	0x0E	1024.25

Current Ranges (CH1-3)

Range	Range ID	Command	Gain
1	1A	0x34	1.07
2	3A	0x04	3.14
3	10A	0x12	10.1
4	30A	0x02	30.6
5	100A	0x01	106
6	100A	0x01	106
7	CT INPUT	0x08	1

Current Ranges (CH4)

Range	Range ID	Command	Gain
1	0.002A	0x74	0.002
2	0.05A	0x04	0.05
3	0.5A	0xD2	0.5
4	1.5A	0x42	1.5
5	10A	0x02	10
6	30A	0x41	30
7	CT INPUT	0x08	1